

EXERCISES AND EXACT SOLUTIONS

Seasonal Crop Production Forecasting

Independent practice, facilitated discussion, and assessment reference

Production Unit	PU-B07-C11
Notebook	V7_A_N06
Version	0.1
Assessment boundary	Learning evidence—not operational authorization

Instructions

Complete Part A before consulting the solutions. State assumptions explicitly. Where evidence is insufficient, write an abstention or escalation response rather than inventing precision.

Exercise 1

Implement rolling-origin validation.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 2

Compare yield-first and direct-production models.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 3

Add a drought-shock stress test.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 4

Explain why import quantities cannot be derived mechanically from the point forecast.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exact solutions and scoring guide

Solution 1

At each historical origin, train only on prior seasons and score the next defined horizon.

Solution 2

Yield-first preserves area $\times$ yield structure; direct models may fit production but obscure components—compare both out of time.

Solution 3

Apply plausible rainfall/yield shocks and report error/interval degradation.

Solution 4

Trade decisions also require stocks, consumption, losses, prices, logistics, policy objectives, uncertainty, and authority.

Scoring rubric

Criterion	Strong evidence	Revision required
Decision framing	Owner, action, horizon, authority named	Model output substituted for decision
Evidence	Concepts, grain, provenance, limits explicit	Data treated as self-explanatory
Uncertainty	Abstention/escalation used appropriately	False precision or hidden limitations
Governance	Human authority and safeguards retained	Authority silently delegated